

Takuya Okawa
(Last updated: Sep 2025)

RESEARCH INTERESTS	Indirect detections of Physics Beyond the Standard Model, physics of the primordial universe, and neutrino physics.	
EMPLOYMENT	<i>IFPU postdoctoral fellow, Scuola Internazionale Superiore di Studi Avanzati (SISSA)</i>	Sep. 2025 -
EDUCATION	<i>Ph.D., Physics, Washington University in St. Louis</i>	Sep. 2020 - Aug. 2025
	Advisor: Francesc Ferrer, P. S. Bhupal Dev <i>B.S., Physics, University of Tokyo</i>	Apr. 2016 - March 2020
EXPERIENCE	<i>Visiting Scholar, Fermilab</i>	July - Dec. 2024
	Advisor: Gordan Krnjaic <i>Visiting Student, Princeton University</i>	Sep. - Nov. 2019
PUBLICATIONS	1. J. H. Buckley, P. S. B. Dev, F. Ferrer, and T. Okawa, "Probing Heavy Axion-like Particles from Massive Stars with X-rays and Gamma Rays," arXiv:2412.21163	
	2. A. Ireland, G. Krnjaic, and T. Okawa, "The Affleck-Dine Curvaton," arXiv:2410.13712	
	3. S. Das, P. S. B. Dev, T. Okawa, and A. Soni, "Old neutron stars as a new probe of relic neutrinos and sterile neutrino dark matter," arXiv:2408.01484	
	4. T. Okawa and A. Y. Chen, "A Model for Pair Production Limit Cycles in Pulsar Magnetospheres", Astrophys.J. 970 (2024) 1, 46	
	5. P. S. B. Dev, F. Ferrer, and T. Okawa, "On the Galactic radio signal from stimulated decay of axion dark matter," JCAP 04 (2024) 045	
AWARDS	<i>Dean's award for Graduate Research Excellence, semifinalist</i>	Aug. 2024
	<i>URA visiting scholar award</i>	April 2024
	<i>Pitt PACC travel award</i>	May 2023
	<i>Ito Foundation U.S.A.-FUTI Scholarship</i>	May 2020
INVITED SEMINARS	<i>University of Winnipeg Theory Seminar (online)</i>	Jan. 2025
	<i>Argonne National Laboratory Theory Seminar</i>	Dec. 2024
	<i>Virginia Tech Center for Neutrino Physics Seminar</i>	Nov. 2024
TALKS	<i>5th EuCAPT Annual Symposium (online)</i>	May 2025
	<i>Particle Physics on the Plains 2024</i>	Nov. 2024
	<i>TeV Particle Astrophysics 2024</i>	Aug. 2024
	<i>TASI 2024</i>	June 2024
	<i>DPF-PHENO 2024</i>	May 2024
	<i>Particle Physics on the Plains 2023</i>	Oct. 2023

	<i>Brookhaven Forum 2023 Advancing Searches for New Physics</i>	Oct. 2023
	<i>N3AS Summer School 2023</i>	July 2023
	<i>Phenomenology 2023 Symposium</i>	May 2023
	<i>APS April meeting 2023</i>	April 2023
	<i>Particle Physics on the Plains 2022 Part 2</i>	Oct. 2022
	<i>XV International Conference on Interconnections between Particle Physics and Cosmology 2022</i>	June 2022
	<i>Particle Physics on the Plains 2022</i>	April 2022
INTERNAL TALKS	<i>Cosmology and astrophysics group meeting, WUSTL</i>	Sep. 2024
	<i>Computational Astro Theory group meeting, WUSTL</i>	Feb. 2024
	<i>Cosmology and astrophysics group meeting, WUSTL</i>	Feb. 2024
	<i>Cosmology and Astrophysics group meeting, WUSTL</i>	Dec. 2023
TEACHING/ DEPARTMENT ACTIVITY	<i>Teaching Assistant</i>	
	Undergraduate course:	
	• Mechanics, PHY 411	Fall 2021
	Graduate course:	
	• Group Theory and Symmetries in Physics, PHY 543	Spring 2023
	• Classical Electrodynamics I, PHY 505	Spring 2022
	Organizer, Student/Postdoc Q&A Session in Theory Seminar	Aug. 2022 - July 2024
	Organizer, Theory Seminar	Aug. 2023 - July 2024
	<i>Supervision</i>	
	Undergraduate student	
	• Robert Ellis	May 2024 - present